

SPPL-199: Min Max DoH Get API technical workflow

Jira Ticket: [SPPL-199: BE | Create technical workflow for Min Max DoH Get API on parameter inputs page including DB details](#) **ACCEPTED**

- i** This page provides comprehensive information about the GET API for Min Max DoH on the Input params page, to view Minimum and Maximum Days on Hand details for the scenario.

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GPS Home / Vanning Rundown Summary / AP TMMK Line3 Aug2025 V1

AP/TMMK Line 3/Aug 2025 V1 Mark Inputs as Ready

Views ><
Search input text
Line Level Inputs
Vanning Center Inputs
Grouping Settings
Grouping
Min Max DoH
Fluctuation Allowance
Simulation
Review
Reports

Min Max DoH See Master Data
Status: **WORK IN PROGRESS**
☐ Apply changes to all groups

Group Name	Min DoH	Max DoH
Group 1 - TMH RAV4 Gas NX HV NX Gas	☒ Apply to all months 12	☒ Apply to all months 18
Group 2 - TMK Camry Highlander	☒ Apply to all months 15	☒ Apply to all months 20

Cancel Save Ready

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Min Max DoH See Master Data
Status: **WORK IN PROGRESS**
☐ Apply changes to all months

Group Name	Min DoH	Max DoH
Group 1 RAV4 Gas NX HV NX Gas	☒ Apply to all months 5	☒ Apply to all months 10
Group 2 Camry Highlander	☒ Apply to all months 15	☒ Apply to all months 20

Monthly overrides 3 months overridden
Adjust Min/Max stock by month. Unchanged months use the group defaults.

Set Min and Max Reset to Defaults 3 months selected Clear selection

Jan 2025	Feb 2025	Mar 2025	Apr 2025	May 2025	Jun 2025	Jul 2025
☒ PRE POPULATED Min (Days) 5 Max (Days) 10	☒ PRE POPULATED Min (Days) 5 Max (Days) 10	☒ PRE POPULATED Min (Days) 5 Max (Days) 10	☒ MANUAL EDIT Min (Days) 5 Max (Days) 10	☒ PRE POPULATED Min (Days) 5 Max (Days) 10	☒ PRE POPULATED Min (Days) 5 Max (Days) 10	☒ PRE POPULATED Min (Days) 5 Max (Days) 10

Cancel Save Ready

Supply Planning GET Min Max DoH API Details:

1. Hosted URL: <https://api.spln.cube<env>.toyota.com/<env>>

2. **API Name:** <HOSTED_URL>/v/1/1/sp/min-max-doh
3. **Path Param:** N/A
4. **Query Param:** ?scenarioId=uniqueScenarioId&userEmail=userEmail
5. **Method:** GET
6. **Authorization:** Bearer token (**required**)
7. **Lambda Name:** spln-<env>-view-min-max-doh-v1-1
8. **API Request Body:** N/A
9. **API Response:**
 - **Success response with status 200:**

```
1 {
2   "scenarioId": "uniqueScenarioId",
3   "data": [
4     {
5       "groupId": "0c1b2a3d-1111-2222-3333-444455556666",
6       "groupName": "Group 1",
7       "vanningCenter": "TMH",
8       "subSeriesList": ["RAV4 Gas", "CAMRY"],
9       "minMaxDoh": {
10        "isUniformAcrossMonths": true,
11        "ranges": [
12          { "year": 2026, "month": 3, "monthName": "Mar", "min": 14, "max": 10 },
13          { "year": 2026, "month": 4, "monthName": "Apr", "min": 14, "max": 10 }
14        ]
15      }
16    },
17    {
18      "groupId": "0d1b2a3d-1111-2222-3333-444455557777",
19      "groupName": "Group 2",
20      "vanningCenter": "TMK",
21      "subSeriesList": ["HighLander", "CAMRY"],
22      "minMaxDoh": {
23        "isUniformAcrossMonths": false,
24        "ranges": [
25          { "year": 2026, "month": 3, "monthName": "Mar", "min": 10, "max": 14 },
26          { "year": 2026, "month": 4, "monthName": "Apr", "min": 11, "max": 10 }
27        ]
28      }
29    }
30  ],
31  "scenarioSteps": {
32    "Line Level Input": {
33      "Model Change Dates": "Completed",
34      "NAMC Allocation Plan": "Completed",
35      "NAMC Production Calendar": "Completed"
36    },
37    "Vanning Center Inputs": {
38      "Shipping Pattern": "Completed",
39      "Vanning Lead Time": "Completed",
40      "TMC Working Day Calendar": "Completed"
41    },
42    "Grouping Settings": {
43      "Grouping": "Completed",
44      "Min Max Stock": "In Progress",
45      "Fluctuation Allowance": "Not Started"
46    },
47    "other": [
48      "Simulation",
```

```

49     "Review",
50     "Reports"
51   ]
52 }
53 }

```

- **Error Response:**

- **any internal server error occurred with status 500:**

```

1  {
2  "errorMessage": "Internal Server Error"
3  }

```

- **any unauthorized error occurred with status 401 (Managed by the AWS API Gateway integrated Custom Authorizer):**

```

1  {
2  "message": "Unauthorized"
3  }

```

- **If request is in-valid with status 400:**

```

1  {
2  "errorMessage": [<"ValidationError: validation error message">]
3  }

```

- **List of possible validation error message:**

1. "ValidationError: scenarioId is required and must be a uuid."
2. "ValidationError: Scenario doesn't exist."
3. "ValidationError: userEmail is required and must be a string."
4. "ValidationError: Invalid userEmail."

10. Table:

- a. **scenarios**
- b. **grouping**
- c. **group_scenario_mapper**
- d. **group_namc_line_series_mapper**
- e. **namc_line_subseries**
- f. **min_max_doh**
- g. **scenario_step_status**

SP Fetch Min Max DoH Data API Workflow:

- **Technical workflow:**

- Description:
 - i. The Supply Planning user logs in to the application.

- ii. After the user provides creates a scenario, opens it and navigates to Min Max DoH, a request is sent to the SP Get Min Max DoH endpoint: **sp/min-max-doh** with parameters: scenarioId=uniqueScenarioId&userEmail=userEmail
- iii. After successful authentication, the **spln-view-min-max-doh-v1-1** lambda is triggered by the SP view scenario Min Max DoH endpoint: **sp/min-max-doh**. If authentication fails, an error with status 401 (*refer: error response-**any unauthorized error occurred with status 401***) will be returned and sent to client.
- iv. After user authentication has been confirmed, the **spln-view-min-max-doh-v1-1** lambda will perform validation on the request payload. If payload is valid will proceed to next step else throw validation error (*refer: error response: **Section if request is in-valid with status 400 and List of possible validation error message***). Below query is used to check if provided scenario is valid:

```
1 select * from supply_planning.scenarios where scenario_id = :scenarioId and  
is_active = true;
```

- v. After the request payload is successfully validated, the **spln-view-min-max-doh-v1-1** lambda execute below queries to get scenario Grouping & minMaxDoh data from the **grouping, group_scenario_mapper, group_namc_line_series_mapper, namc_line_subseries & min_max_doh** tables:

```
1  /*  
2   Purpose:  
3   - Fetch all active groups for a scenario ($1)  
4   - Fetch each group's subSeries (1 row per subSeries)  
5   - Fetch each group's DoH min/max per month/year (1 row per month)  
6  */  
7  SELECT  
8      gp.group_id                AS "groupId",  
9      sgm.group_name             AS "groupName",  
10     gp.vanning_center          AS "vanningCenter",  
11     nls.sub_series_description AS "subSeries"  
12 FROM supply_planning.group_scenario_mapper sgm  
13 JOIN supply_planning.grouping gp  
14     ON gp.group_id = sgm.group_id  
15     AND gp.is_active = TRUE  
16 JOIN supply_planning.group_namc_line_series_mapper gnls  
17     ON gnls.group_id = gp.group_id  
18     AND gnls.is_active = TRUE  
19 JOIN supply_planning.namc_line_subseries nls  
20     ON nls.namc_line_series_id = gnls.namc_line_series_id  
21     AND nls.is_active = TRUE  
22 WHERE sgm.scenario_id = ${scenarioId}::uuid  
23     AND sgm.is_active = TRUE;  
24  
25 /* Fetch min, max, month, year by the provided scenarioId for the associated  
26    groups (groupId+groupName) */  
27 SELECT  
28     gp.group_id        AS "groupId",  
29     sgm.group_name     AS "groupName",  
30     gp.vanning_center  AS "vanningCenter",  
31     mmd.year           AS "year",  
32     mmd.month          AS "month",
```

```

32     mmd.min_doh         AS "minValue",
33     mmd.max_doh         AS "maxValue"
34 FROM supply_planning.group_scenario_mapper sgm
35 JOIN supply_planning.grouping gp
36 ON gp.group_id = sgm.group_id
37 AND gp.is_active = TRUE
38 JOIN supply_planning.min_max_doh mmd
39 ON mmd.group_id = gp.group_id
40 AND mmd.scenario_id = sgm.scenario_id
41 WHERE sgm.scenario_id = ${scenarioId}::uuid
42 AND sgm.is_active = TRUE;

```

vi. Below query is used to fetch view scenario menu data: ref ([SPPL-162: View Scenario Menu technical workflow document](#))

```

1 | select * from supply_planning.scenario_step_status where scenario_id =
   | :scenarioId;

```

vii. The retrieved data is formatted as shown below before returning the response to the client with a status code of 200:

1. If the Min Max DoH data for scenario is available:

```

1  {
2    "scenarioId": "uniqueScenarioId",
3    "data": [
4      {
5        "groupId": "0c1b2a3d-1111-2222-3333-444455556666",
6        "groupName": "Group 1",
7        "vanningCenter": "TMH",
8        "subSeriesList": ["RAV4 Gas", "CAMRY"],
9        "minMaxDoh": {
10         "isUniformAcrossMonths": true,
11         "ranges": [
12           { "year": 2026, "month": 3, "monthName": "Mar", "min": 14, "max":
13             10 },
14           { "year": 2026, "month": 4, "monthName": "Apr", "min": 14, "max":
15             10 }
16         ]
17       },
18       {
19         "groupId": "0d1b2a3d-1111-2222-3333-444455557777",
20         "groupName": "Group 2",
21         "vanningCenter": "TMK",
22         "subSeriesList": ["HighLander", "CAMRY"],
23         "minMaxDoh": {
24           "isUniformAcrossMonths": false,
25           "ranges": [
26             { "year": 2026, "month": 3, "monthName": "Mar", "min": 10, "max":
27               14 },
28             { "year": 2026, "month": 4, "monthName": "Apr", "min": 11, "max":
29               10 }
30           ]
31         },
32         "scenarioSteps": {
33           "Line Level Input": {
34             "Model Change Dates": "Completed",
35             "NAMC Allocation Plan": "Completed",
36             "NAMC Production Calendar": "Completed"
37           },
38           "Vanning Center Inputs": {

```

```

38     "Shipping Pattern": "Completed",
39     "Vanning Lead Time": "Completed",
40     "TMC Working Day Calendar": "Completed"
41 },
42 "Grouping Settings": {
43     "Grouping": "Completed",
44     "Min Max Stock": "In Progress",
45     "Fluctuation Allowance": "Not Started"
46 },
47 "other": [
48     "Simulation",
49     "Review",
50     "Reports"
51 ]
52 }
53 }

```

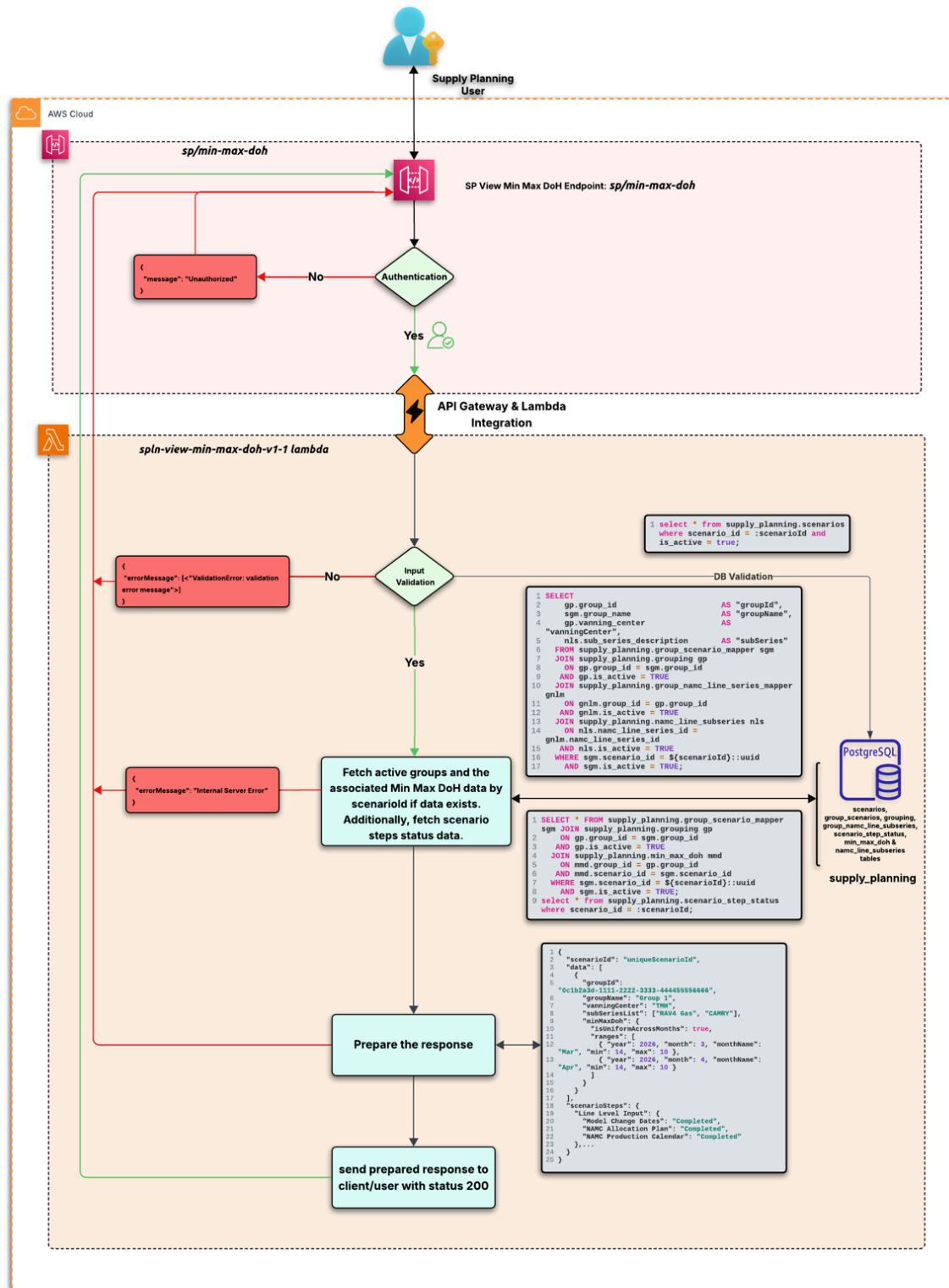
2. If Min Max DoH data is not available:

```

1  {
2    "scenarioId": "uniqueScenarioId",
3    "data": [],
4    "scenarioSteps": {
5      "Line Level Input": {
6        "Model Change Dates": "Completed",
7        "NAMC Allocation Plan": "Completed",
8        "NAMC Production Calendar": "Completed"
9      },
10     "Vanning Center Inputs": {
11       "Shipping Pattern": "Completed",
12       "Vanning Lead Time": "Completed",
13       "TMC Working Day Calendar": "Completed"
14     },
15     "Grouping Settings": {
16       "Grouping": "Completed",
17       "Min Max Stock": "In Progress",
18       "Fluctuation Allowance": "Not Started"
19     },
20     "other": [
21       "Simulation",
22       "Review",
23       "Reports"
24     ]
25   }
26 }

```

viii. If there is an error while fetching the data, an “Internal Server Error” with a status code of 500(refer: error response-**any internal server error occurred with status 500**) will be returned. Otherwise, a successful response with a status code of 200 will be sent.



SP Min Max DoH Get API Technical workflow

Note:

We will be utilizing this API for both to fetch file data and previous rundown data.

Queries: